Colorscope, Inc.

Introduction

Andrew Cha, the founder of Colorscope, Inc., a small, vibrant firm in the graphic arts industry, had seen his business change dramatically over the years. The rapid development of such technologies as desktop publishing and the World Wide Web as well as the consolidation of several major players within the industry had radically altered his company's relative positioning on the competitive landscape. Preparing to celebrate the company's twentieth anniversary in March 1996, Cha pondered the issues involved in moving Colorscope ahead.

Company History

Born in Anhui, China in 1938, Andrew Cha immigrated to the United States in 1967 to seek a better life. Originally planning to settle in New York City, where he would pursue his craft as a painter and his wife would attend New York University, his funds ran out in Los Angeles, forcing him to work as a cook and busboy in a downtown Chinese restaurant. Through fortune and hard work, however, Cha eventually found jobs that took advantage of his artistic skills in draftmanship and photography; a succession of promotions within one graphic arts company convinced him that his abilities would enable him to start his own business. Founded on March 1, 1976, Colorscope Inc. was established as a special-effects photography laboratory serving local advertising agencies in southern California.

As Cha's reputation grew, so did the business. Sales increased steadily over the years, peaking in 1988 at $5 million dollars. The company served agency giants such as Saatchi & Saatchi, Grey Advertising, and J. Walter Thompson and large retailing and entertainment companies such as The Walt Disney Company and R. H. Macy & Co. To improve service to these customers, Cha invested in expensive proprietary computer equipment to continue providing ever more complicated print special effects.

During 1988, Cha was approached by R. R. Donnelley & Sons Co. about a possible acquisition. Donnelley, the largest printer in the world with roughly $4.3 billion in sales at the time, was interested in acquiring Colorscope for approximately $10 million. The interest in Colorscope was twofold. First, Cha had built solid relationships with highly valuable print and pre-press buyers in the marketplace. Every pre-press dollar he sold was worth several more in printing. By owning

Joseph Cha, HBS MBA class of 1996 and Assistant Professor V.G. Narayanan prepared this case as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

Cha's pre-press business and employing him as a sales consultant, Donnelley hoped to secure large print contracts, which at the time were still subject to open bidding. Second, Cha's operation was considered one of the most efficient in the business. Donnelley employees had previously visited his operation and modeled some of his workflows, adapting them into the design of one of their own pre-press facilities. As a result, Donnelley considered Cha's business processes as well as his training methods an operational advantage they could leverage to other pre-press facilities in their network of operations across the country.

After considering his options and his belief in the potential of the business, however, Cha grew dissatisfied with several of the contingency and non-compete clauses built into the agreement, and eventually decided against the sale of his company. The timing of this decision proved costly. While serving his existing base of high-margin clients, Cha ignored certain trends in the business, particularly the price pressures brought on by cheaper PC- and Mac-based microcomputers. As these devices, equipped with increasingly sophisticated page layout and color correction software, proliferated and increased in functionality, small ad agencies and print shops began to take pieces of business away from larger graphic art companies like Colorscope. Cha, however, had felt protected from the trend by the strong personal relationships he had built with key clients over his career.

Nevertheless, by 1990, technology and the pace of change in the desktop publishing industry forced significant changes in his business. The first impact was on pricing. Although he emphasized the quality and reliability of his work, market pressure forced him to reduce his own basic prices. which previously had held up against industry trends. (See **Exhibit 1**). This, however, proved to be insufficient. In May 1994, his largest account, representing about 80% of his business, announced that it was purchasing its own graphic design and production equipment, replacing Colorscope with an internal group. The process was to be phased in over the following year. After losing his most significant and long-term client, Cha thought that to rebuild the business he had to reevaluate the industry, his company's position in the pre-press segment, its pricing policy, and its operations.

The Pre-press Production Process

Although technology dramatically changed the means by which production was conducted as well the corresponding values to each phase, the basic process for print material, known in the industry as pre-press or color separations, remained essentially the same over the past 20 years. (See **Exhibit 2**). A content provider, such as a magazine or direct mail cataloger, designed and laid out a "book" or "project" for distribution. Once the book's layout was approved, a photographer captured and developed the images, received approvals from the client, and sent them to the pre-press house or "color separator," in this case Colorscope. Once in production, images were processed or digitized via laser scanner and compiled with text and other graphics to form a master file for the printer.

During this process, the magazine or direct mail client saw iterations, or proofs, of their "book" with digital and conventional proofing devices. At these intervals, the clients could ask for changes, ranging from simple price and copy adjustments to sophisticated special effects, adjusting colors or clearing blemishes in products and people. A very important qualitative component of the separator's task was understanding the product's desired "look and feel" and translating the direction the client desired into the actual images on each page. Typically, the pre-press house charged a base rate for digitizing, assembling, and proofing each page, with an additional fee for the special effects. When the "books" were ultimately produced on paper, the images were filed and stored in the separator's database for future use.

After the project gained final approval, Colorscope sent the "master book," or file, to the printer either electronically or by large sheets of four-color master film. At this point, the separator had converted all of the client's information, digital text, graphics, and photographs (described in a postscript or dpi format) into a printer-acceptable (line screen) format. Once at the printer, the film or information was converted to physical plates of various materials (metal, alloys, or plastics) specific to the size of the item and the number to be printed. These plates were mounted on drums, and the information became imprinted via an offset or gravure printing process with multiple ink types onto a wide ranging array of papers depending again on the size of the item and order size of the project. After the printer finished the pieces (i.e., cutting, binding, and addressing), the final "books" were ready for shipment to the magazine or direct mailer's subscribers.

Industry Dynamics

The overall market for commercial printing services in the United States topped $66 billion in 1995. Because of the highly diverse range of printed material produced, companies in the print industry tended to specialize in market subsets like greeting cards, business forms, financial reports, newspapers and newspaper inserts, magazines, direct mail catalogs, coupons, directories, etc. This evolved specialization led to a highly fragmented competitive landscape, where most companies served a few primary clients, highlighting an operational expertise in an area narrow enough to discourage other competitors. Thus, if a client planned an advertising campaign that involved several media, e.g., point-of-sale stands, packaging, and direct mail catalogs, it might use several different printers and print distributors for the different products. As a result, in the commercial printing business for catalogs, there were only a handful of printers with the necessary capacity and marketing strength to compete effectively for large print customers. (See **Exhibit 3**.)

The pre-press market mirrored the print industry but on a smaller scale. Given the highly fragmented nature of the business as well as a paucity of published information about actual pre-press sales on a national level, it was difficult to quantify the exact market size for pre-press services. Because the typical catalog pre-press job represented approximately 10% to 20% of the printing price charged to the client, however, the U.S. pre-press market in 1995 could have been as large as $6 billion (see **Exhibit 4**). Specialization among competitors, similar to the print business, was common. Although a client might choose a single pre-press firm to house all of its images to interface with several printers, many larger graphics customers employed several separators to handle different projects if they had different expertise with specific print products. Those clients would, for example, use separators specialized in video box (i.e., packaging for VHS video cassettes), posters, or printed books.

For the individual pre-press firm the market had drastically changed. Thus Colorscope's previous position as a high quality, high service player appeared unsustainable in a marketplace full of service providers that claimed the same quality at lower prices (see **Exhibit 5)**. While in the past prosperous relationships could last several years, with customers consistently able and willing to pay for top quality separations, current technology blurred the clear distinctions in quality of the actual film output. As a result, some previously loyal customers looked at pre-press services as more of a commodity product; correspondingly, personal relationships alone no longer seemed to ensure the livelihood of Cha's business. As more pre-press houses bought desktop equipment and lowered their prices, customers in the catalog arena defected to even lower-cost providers. By 1995, the base price for a typical direct mail catalog had stabilized around $500-$600 per page, roughly half the rates charged only five years before. Given that the basic scanning and proofing functions of a pre-press house could be easily replicated on a smaller scale with minimal investment,[1] and given the significant overcapacity in the industry, Cha knew that the downward pressure on prices was likely to continue.

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[1] Local service bureaus could scan color film, layout pages, and output printer specified film with a minimal capital investment of less than $100,000.

Direct Competition

Although the number of larger direct mail clients had remained flat for several years, the competition for them was intense. Cha's competitors were no longer other local craftsmen with conventional cameras and artistic skills; rather, they came in three main types. First were larger, more technically savvy printing companies with professional salespeople pushing bundled pricing, integrating pre-press services with printing in a single package. Rivals here consisted of national printers with multimillion dollar and billion dollar-plus revenues like R. R. Donnelley & Sons Co. and Quad Graphics, which had integrated backward into pre-press services over the last decade. Another significant rival type was represented by the horizontally integrated national pre-press houses or "trade shops" such as American Color and Wace/Techtron—highly entrenched, multi-million dollar pre-press service providers backed by national sales networks of service professionals and multiple physical plant locations across the United States. These companies competed in several different submarkets beyond catalogs, e.g., inserts, comic syndications, and coupons. A third rival type comprised other standalone firms that competed with loose affiliations to other printers or advertising agencies, or that literally set up shop next door to their largest accounts to fend off potential competitors. Cha currently lacked a sales infrastructure similar to that of these competitor types, however.

Work Flow Organization at Colorscope

A "job" at Colorscope began when the customer placed an order. Customer Service representatives interacted with the customer on the phone and recorded the job specification details. Each order was "owned" by a particular representative, who, based on the specifications, did a "job preparation." A separate "job bag" was opened for each set of four pages for the order. The template of the job was created by physically cutting and pasting text, graphics, and photographs; extensive markings on the template specified the changes in font, color, shading, and layout. The next step in the production process was scanning, whereby the pictures were digitized and output as a computer file. Colorscope had three laser scanners.

The following step was assembly, performed on nine high-end, souped-up Macintosh computers, each with 256MB RAM and oversized computer terminals. The computers were networked and hooked up to the scanners, output devices, and a powerful file server with 40 gigabytes memory that contained archives of optical images. Operators worked on the computers composing the "job" with scanned images and text input from the keyboard. At this stage the operators changed colors and shades of the scanned picture to the exact specifications the customer demanded. Once a job was fully assembled, it was output on one of two high-end output devices. The output was a large sheet of four-color film that was then developed.

The "job" then flowed to Quality Control (QC) for proofing. Proofing involved comparing the hardcopy output with customer specifications. Reworks were initiated at this stage. QC might, for instance, require the job to be rescanned if it determined that the original scanning was flawed. Then the rescanned image would then have to be reassembled, re-output, and pass QC all over again. Once a job passed QC, it was shipped to the customer's printer either on a computer disk or, more usually, on film.

Colorscope's operators were cross-trained and could work on any stage of the production process. Work flow and production procedures were standardized but not documented. Colorscope relied instead on the institutional knowledge of its employees and frequent supervision by Andy Cha to maintain and improve operational efficiency.

The Future

By the spring of 1996, Cha realized that Colorscope had to capitalize on its biggest assets, its employees, who were all well trained and worked effectively as a team to meet deadlines. (See **Exhibits 6** and **7**.) The short-term strategy was to increase marketing efforts to drum up new business for the lean months that preceded the huge rush of orders to do pre-press for catalogs in the fall season before holiday shopping started. **Exhibit 8** gives details of jobs completed in June 1996 and revenue generated from each customer. Revenue per page, however, was unlikely to improve due to competitive pressures. Cost containment and improving operational efficiency were, therefore, critical, particularly in reducing the amount of rework. This effort required the cooperation of its workers, and Cha was considering sharing the gains of such improvement with its employees. With this objective in mind, Colorscope began tracking hours spent on rework, which was broken down into hours spent on rework initiated by customer due to change in specifications, and rework caused by errors in-house. Colorscope compensated its line workers on an hourly wage basis. To keep track of hours worked, employees logged the hours spent on various jobs into a centralized computer from remote terminals. (**Exhibit 9** gives the hours spent at different workstations by different jobs in June 1996.) So tracking rework hours was fairly straightforward; employees recorded both types of rework hours separately for each job. (**Exhibit 10** gives the rework hours recorded during June 1996.)

Another area for improvement was product pricing. At present Colorscope quoted more or less the same per-page price for different customers, plus additional charges for special effects. Yet different customers placed different demands on organizational resources, and this was not appropriately reflected in the price charged. However, Colorscope could not afford expensive accounting systems or to hire consultants to design a state-of-the-art activity-based cost system. **Exhibit 11** gives selected financial information while **Exhibit 12** gives materials expense, broken down by jobs, for the month of June 1996.

Questions for the Future

As Andrew Cha anticipated the celebration of Colorscope's twentieth year in business, he jotted down three questions that summed up the areas of improvement he felt were needed. How could Colorscope improve its operations? How could it change its pricing strategy? What accounting and control system should his company install?

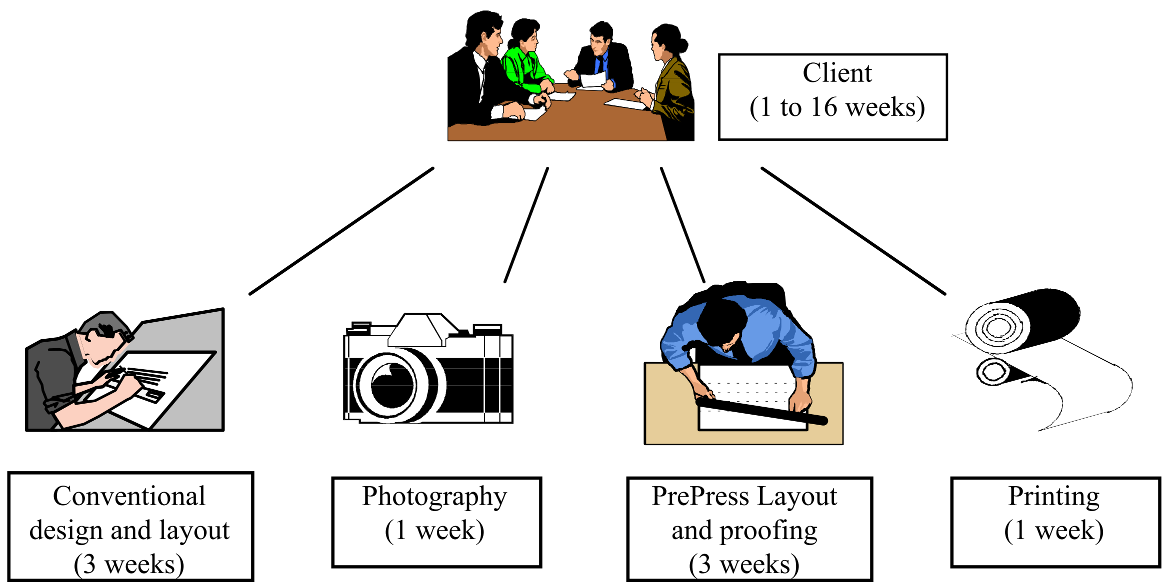
Exhibit 1 Price trend of color separations to scan, assemble, proof and output printer specified film for a hypothetical direct mail catalog page with one image

Price/Page of work ($):	1991	1992	1993	1994	1995
Colorscope	1,100	900	750	650	600
Service-Bureau[2]	400	425	450	475	500

[2] Service -Bureaus such as Kinko's and Alphagraphics typically do jobs that are cheaper, and involve fewer and/or smaller pages.

Exhibit 2 How technology has changed the Prepress Process

Up until the early 1990s, before the dawn of desktop publishing, print production "end to end" for a hypothetical book required (excluding client approval times for merchandising) approximately 8 weeks on average. The average catalog client would hire different vendors to provide each service and coordinate the production process. The diagram below outlines how four separate companies would act in concert to produce a typical book.



With the advent of desktop publishing, print production "end to end" times shrank on average by two weeks. Savings come mainly from time saved in the artists' and production technicians' capacity to make changes quickly and easily. Digital photography has potential to shrink photography times even further.

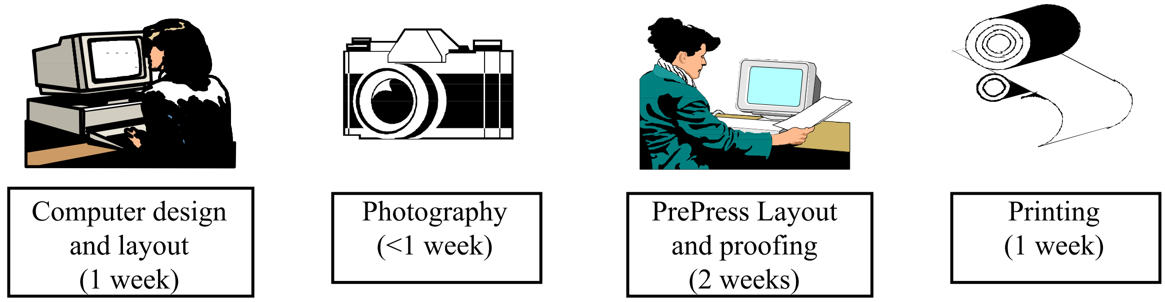


Exhibit 3 Top 25 American Printers and Specialization's

Rank	Company	Sales (Million $US)	Primary Business
1	R. R. Donnelley & Sons Co.	4,800	BK, CAT, DIR, FIN, INS, PUB
2	Hallmark Cards, Inc.	3,800	GC, SPC
3	Moore Corp. Ltd.	2,401	BF, SPC
4	Quebecor Printing, Inc.	2,116	BK, CAT, COM, INS, PUB
5	American Greetings Corp.	1,870	GC, SPC
6	Deluxe Corporation	1,748	BF, GC
7	World Color Press, Inc.	970	CAT, COM, DIR, PUB
8	Banta Corporation	811	BK, CAT, COM, PUB, SPC
9	Treasure Chest Advertising Co.	807	INS, SPC
10	Quad/Graphics, Inc.	801	CAT, COM, INS, PUB
11	Standard Register	767	BF, COM
12	UARCO, Inc.	599	BF, SPC
13	Wallace Computer Services, Inc.	588	BF, COM, SPC
14	American Business Products	563	BF, BK, PKG, SPC
15	AT&T Systemedia Group	560	BF
16	Gibson Greetings, Inc.	549	GC, SPC
17	Valassis Communications, Inc.	543	INS, SPC
18	John H. Harland Co.	521	BF, SPC
19	Taylor Corporation	488	COM
20	Sullivan Communications	475	BK, INS, PUB
21	Ringier America, Inc.	471	BK, CAT, INS, PUB
22	Reynolds & Reynolds	426	BF
23	Transcontinental Printing, Inc.	416	CAT, COM, INS, PUB
24	Western Publishing Co.	398	BK, CAT, DIR, PUB, SPC
25	Brown Printing Co.	384	CAT, INS, PUB

Business Category Abbreviations:

BF	Business Forms	DIR	Directories	PKG	Packaging
BK	Book Publishing	FIN	Financial	PUB	Publications
CAT	Catalogs	GC	Greeting Cards	SPC	Specialty Printing
COM	Commercial	INS	Inserts		

Source: American Printer, 1995

Exhibit 4 Time/Cost Breakdown for the Catalog Producer

Direct mail catalogers constituted over 90% of Colorscope's customers. To produce an average project run, in this example, 1 million copies of a 52-page catalog, costs would break down as follows:

Design (3 weeks), writing copy, producing layouts, gaining management approvals.	Photography (2 weeks), 200-300 images, studio and location shots.	Prepress (2 weeks), 2 rounds of proofs to the project manager.	Printing (3 days), cutting, binding, addressing for national distribution	Distribution (2-3 weeks), second class bulk rate post.
$10,000	$50,000	$30,000	$1,300,000	$120,000

Exhibit 5 Top American Pre-press Competitors
(does not include printers that have integrated forward)

Company	**Sales (M$US)**	**Employees**	**Facilities (sites)**	**Primary Business**
WACE USA	200	1,850	20	magazines, direct mail, packaging
Applied Graphics Technologies	130	1,060	13	magazines, direct mail, packaging
Black Dot Group	109	900	12	mags, dir. mail, packaging, books
Schawk, Inc.	104	729	12	mags, dir. mail, packaging, agency
American Color	76	840	16	mags, dir. mail, pkg, agncy, newsp
Intaglio Vivi-Color	60	450	13	packaging
Enteron Group	56	400	4	mags, dir. mail, pkg, agncy, books
Kwik International	41	110	1	packaging, agency
Color Associates	37	390	1	packaging, publishing, catalogs
TSI Graphics	27	313	4	mags, dir. mail, packaging, books
Blanks Color Imaging	24	230	1	mags, dir. mail, packaging, agency
Kreber Graphics, Inc.	22	165	2	mags, dir. mail, pkg, agncy, retail

Exhibit 6 Key Personnel at Colorscope

Andrew ("Andy") Cha - founder and president, since 1976. In charge of operations and sales. Plays "bad cop" on the shop floor to make sure productivity stays ultra high. Major contribution to the business is his experience base which he leverages to close new sales.

Agatha ("Aggie") Cha - VP Marketing, since 1976. Works with her husband to nurture all client relationships and customer and prospect needs. Functions as head of human resources, making all hiring, firing, and salary recommendations. Also keeps the pulse of workers, keeping track of birthdays, anniversaries, etc.

Joe Cha son of Andrew and Agatha, HBS MBA 1996; Stanford BA in Economics 1991 and MA in Sociology 1992. Worked at Colorscope in marketing during 1993-94, bringing in several new customers; would be spending the summer of 1996 in improving operations and customer relations before moving to Shanghai to pursue a career in consulting.

John Gibson - controller, since 1981. Manages working capital, payroll, and all financing necessary to fund the business. Actually lives with the Cha family.

Ruth Fukushima - head of customer service, since 1980. Manages quality control and customer satisfaction while playing "good cop" on the shop floor to keep operators from burning out under Andrew Cha's management. Organizes company picnics, parties, and other celebrations.

Don Chin - technology director, since 1992. Manages data servers and keeps Andrew Cha on top of new technologies. Handles all new software training issues and is general fix-it man for all hardware problems. A technology whiz and Joe Cha's friend from their undergraduate days together at Stanford, Chin also helps to increase customers' comfort levels with new platforms and technologies.

Exhibit 7 Organization Chart

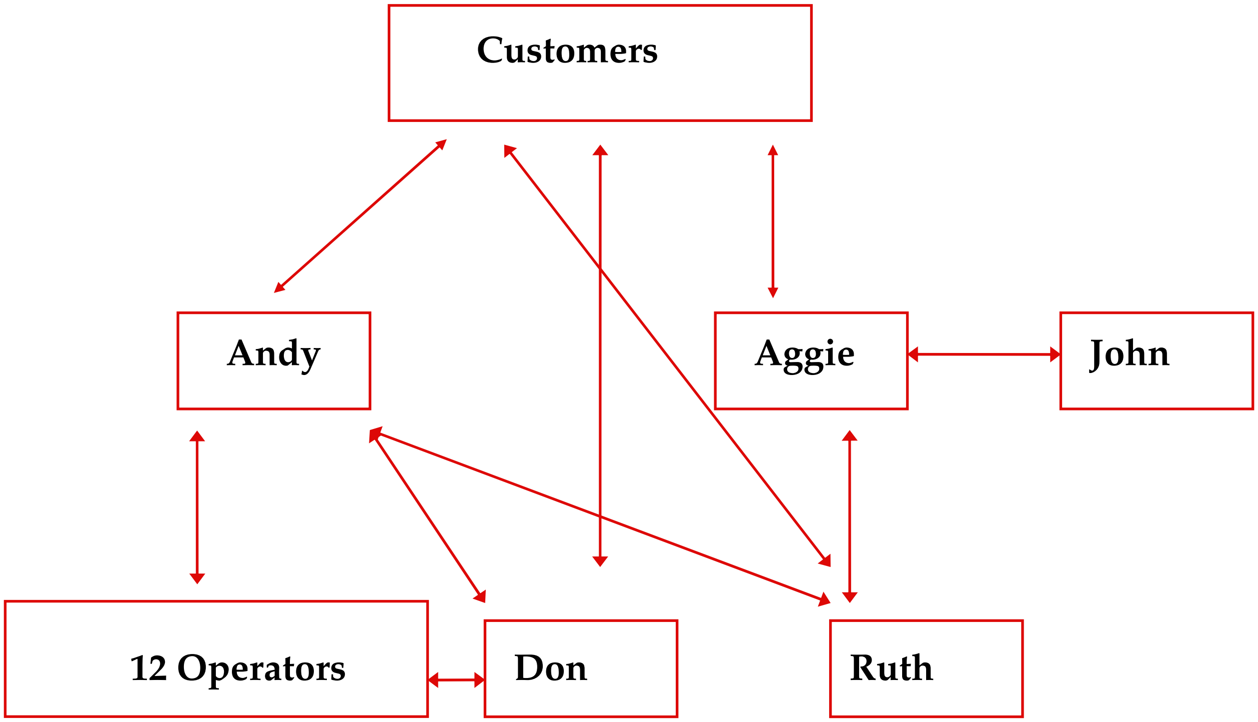


Exhibit 8 Jobs completed in June 1996*

Job #	Customer #	Pages	Revenue
61001	10	16	$ 9,600
61002	10	16	9,600
61003	10	32	23,000
61101	11	16	12,000
61102	11	16	11,000
61201	12	16	11,000
61202	12	32	23,000
61203	12	32	22,000
61204	12	32	20,000
61301	13	128	50,000
61401	14	16	7,800
61402	14	16	8,000
61403	14	16	8,000
61404	14	16	9,000
61405	14	16	9,800
61501	15	16	11,000
61502	15	16	11,000
61601	16	32	20,000
61602	16	4	2,000
61603	16	4	1,400
61701	17	16	8,000
61702	17	16	10,000
61801	18	4	4,000
61901	19	4	2,000
61902	19	16	12,000
62001	20	1	0
Total		**545**	**$315,200**

* All figures are disguised

Exhibit 9 Hours clocked at different workstations in June 1996*

JOB #	Job Preparation	Scanning	Assembly	Output	Quality Control	Total
61001	4	32	42	8	7	**93**
61002	3	24	38	8	8	**81**
61003	7	40	75	16	8	**146**
61101	4	16	30	4	4	**58**
61102	4	16	28	4	4	**56**
61201	4	16	32	4	6	**62**
61202	7	32	58	8	5	**110**
61203	6	34	64	8	6	**118**
61204	6	30	58	8	8	**110**
61301	15	130	250	32	30	**457**
61401	5	14	32	4	4	**59**
61402	4	19	32	8	7	**70**
61403	4	20	34	4	3	**65**
61404	4	22	36	4	5	**71**
61405	4	20	36	4	4	**68**
61501	4	21	39	4	4	**72**
61502	4	20	40	8	7	**79**
61601	7	26	60	8	9	**110**
61602	2	5	10	1	1	**19**
61603	2	5	11	2	1	**21**
61701	4	20	39	4	3	**70**
61702	4	20	41	4	5	**74**
61801	1	5	11	2	1	**20**
61901	2	5	12	1	1	**21**
61902	5	19	42	4	5	**75**
62001	1	1	2	1	1	**6**
Idle Time	43	28	128	37	13	**249**
Capacity	**160**	**640**	**1280**	**200**	**160**	**2440**

Hours clocked in different work stations include rework hours given in **Exhibit 10.**

* All figures are disguised

Exhibit 10 Rework hours*

Rework due to change in specifications by customer

JOB #	Job Preparation	Scanning	Assembly	Output	Quality Control	Total
61001	0	16	10	4	2	**32**
61002	0	8	6	4	3	**21**
61301	2	5	10	2	2	**21**
61502	1	4	8	1	0	**14**
61801	0	1	3	1	0	**5**
61901	1	1	4	0	0	**6**
Total	**4**	**35**	**41**	**12**	**7**	**99**

Quality Control initiated rework of house errors

JOB #	Job Preparation	Scanning	Assembly	Output	Quality Control	Total
61301	1	3	4	1	1	**10**
61402	0	9	16	2	2	**29**
61403	0	10	14	2	1	**27**
61603	1	3	3	1	0	**8**
Total	**2**	**25**	**37**	**6**	**4**	**74**

In all four jobs that were subsequently reworked because Quality Control initiated rework, the original defects were introduced in the scanning stage of the operation. However, when a job is rescanned, assembly, output, and quality control all have to be redone.

* All figures are disguised

Exhibit 11 Selected Financial Information for June 1996*

Description	Job Preparation	Scanning	Assembly	Output	Quality Control	Idle	Total
Wages	$8,000	$32,000	$64,000	$10,000	$11,000		**$125,000**
Depreciation	$500	$25,000	$10,000	$14,000	$500		**$50,000**
Rent							**$30,000**
Others							**$20,000**
Total Overhead							**$225,000**
Floor Space in sq. ft.	1000	1000	4000	2000	500	6,500	**15,000**

Exhibit 12 Materials expense in June 1996*

Job #	Total Materials expense[3]	Customer initiated rework	Correction of house error
61001	$ 5,400	$2,700	
61002	3,500	1,100	
61003	4,500		
61101	1,800		
61102	1,500		
61201	1,500		
61202	3,300		
61203	3,400		
61204	3,200		
61301	13,000	1,000	$1,000
61401	1,800		
61402	3,100		1,000
61403	3,900		1,000
61404	2,100		
61405	2000		
61501	2200		
61502	3,600	1,500	
61601	3,300		
61602	600		
61603	1,000		500
61701	2100		
61702	2,500		
61801	1,600	1,000	
61901	1,700	1,000	
61902	2,200		
62001	200		
Total	**$75,000**	**$8,300**	**$3,500**

* All figures are disguised
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[3] Includes materials for rework.